Macroeconomics I — MPE Insper

Practice list: Consumption, Saving and Labor Markets — qualifier level (Listas 3–4)

Instructions. Multiple choice: mark the single correct option (no penalty for wrong answers). *True/False*: every 2 wrong items cancel 1 right one (score = $\max\{0, \text{right} - \text{wrong}/2\}\times$ item value); blank items do not count. Discursive questions: show every step. Numbers are illustrative and chosen to give clean answers.

Block 1 — Consumption, present bias and interest rates

(Kurlat chs. 6, 9; Romer ch. 8 • 25 points)

Question 1. (10 points) Objective items.

a) (3 points) A household lives 40 periods ($t = 0, \ldots, 39$) with CRRA utility, $\sigma = 2$, $\beta = 0.97$, faces $r = 4\%$ and has total wealth W (financial plus human, in present value at $t = 0$). The share of W consumed at $t = 0$ is:

A. 3.42%

B. 4.55%

C. 2.50%

D. 3.85%

b) (3 points) *Consumption × general equilibrium.* A two-period endowment economy with a representative household: $y_2/y_1 = 1.03$, $\beta = 0.97$, CRRA with $\sigma = 2$. The equilibrium real interest rate is:

A. 3.09%

B. 6.19%

C. 9.37%

D. 3.00%

c) (2 points, T or F) *Consumption × credit markets.* A household strictly inside the region where it neither borrows nor lends (borrowing rate above the deposit rate) starts saving after a small rise in the deposit rate.

d) (2 points, T or F) *Consumption × fiscal policy.* A household at a binding borrowing limit receives a lump-sum tax cut of 100 today, financed by a tax increase of $100(1 + r)$ next period. If the limit still binds after the cut, current consumption rises by exactly 100.

Question 2. (15 points) Discursive. *Consumption × behavioral economics × general equilibrium.* Three periods $t = 0, 1, 2$; in (a)–(c) the interest rate is zero and the household has wealth W at date 0. The self at date t maximizes $u(c_t) + \beta \sum_{s>t} u(c_s)$ with $u(c) = c^{1-\sigma}/(1-\sigma)$ ($\ln c$ if $\sigma = 1$) and $0 < \beta < 1$.

a) (3 points) Find the plan chosen with commitment (self 0 chooses all three consumptions), by a naive self 0 (who believes future selves will follow its plan), and by a sophisticated self 0 (who anticipates self 1's choice).

Solution:

b) (3 points) Determine when the sophisticated self 0 consumes more, less or the same as the naive one, and explain the mechanism.

Solution:

c) (3 points) What share of W would self 0 pay for an illiquid account that commits future consumption? Find a formula, its limits as $\beta \to 1$ and as $\sigma \to \infty$, and compute it for $\sigma = 2$, $\beta = 1/2$.

Solution:

d) (3 points) Now every household receives y at each date and households are identical and sophisticated; they can borrow and lend at r_{01} (between dates 0 and 1) and r_{12}. Find the equilibrium interest rates, show whether they depend on σ, and draw them against $\beta \in [0.4, 1]$.

Solution:

e) (3 points) *Parameter analysis.* How do the value of commitment and the equilibrium rates depend on β and σ? Prove the signs you can, fill the value of commitment for $\sigma \in \{1, 1.5, 2, 5\}$ and

$\beta \in \{0.5, 0.7, 0.9\}$, and read the results.

Solution:

Block 2 — Labor supply, taxes, transfers and search

(Kurlat ch. 7; Romer ch. 10 • 25 points)

Question 3. (10 points) Objective items.

a) (3 points) *Labor × consumption smoothing.* Per-period utility $\ln c - \psi \frac{h^{1+1/\varepsilon}}{1+1/\varepsilon}$, $\varepsilon = 0.5$; 40 periods with $\beta(1+r) = 1$ and equal wages. The wage rises 10% in one period only. To first order, hours in that period rise by:

A. 4.9%

B. 5.0%

C. 0%

D. 3.3%

b) (3 points) *Search × wage setting.* In the steady-state search model with $f = \mu\theta^{\alpha}$, $\alpha = 0.5$, free entry $(y-w)q(\theta) = \kappa(r+s)$, the wage is rigid at $w = 0.95$ while productivity $y = 1$ moves. The elasticity of tightness θ with respect to y is:

A. 3.7

B. 40

C. 20

D. 2.0

c) (2 points, T or F) *Search × efficiency.* With a general constant-returns matching function, the search equilibrium is efficient iff the workers' share of the surplus equals the elasticity of matches with respect to unemployment, evaluated at the equilibrium tightness.

d) (2 points, T or F) *Labor × transfers.* With $u = \ln c + \alpha \ln(1-h)$ and a lump-sum transfer $T > 0$ that each household takes as given, an individual household's hours do not respond to its after-tax wage.

Question 4. (15 points) Discursive. *Labor supply × taxes and transfers × welfare.* Households have $u = \ln c + \alpha \ln(1-h)$ with one unit of time; a linear technology pays $w = 1$ per hour; labor income is taxed at τ and the revenue is returned as a lump-sum transfer T that each household takes as given. Baseline $\alpha = 0.8$.

a) (3 points) Derive individual labor supply given (τ, T). Show what the tax does when $T = 0$, and find equilibrium hours $h(\tau)$ when the budget balances.

Solution:

b) (3 points) Find the tax rate that maximizes revenue (the size of the transfer program), prove that revenue is single-peaked, and draw revenue against τ.

Solution:

c) (3 points) Derive the welfare cost of the program in consumption-equivalent terms (the share of consumption at $\tau = 0$ the household would give up to avoid the program), prove that it is zero to first order at $\tau = 0$ and increasing, and compute it at $\tau = 0.4$, 0.5, 0.6.

Solution:

d) (3 points) At $\tau = 0.4$, compute the Marshallian and Hicksian elasticities of individual hours with respect to the after-tax wage, the balanced-budget elasticity $d\ln h / d\ln(1-\tau)$, and the Frisch elasticity. Which one does Prescott's experiment use?

Solution:

e) (3 points) *Parameter analysis.* How do τ^*, hours at $\tau = 0.5$ and the welfare cost at $\tau = 0.5$

depend on α? Prove the signs you can, give the limits of τ^*, fill the table for $\alpha \in \{0.4, 0.8, 1.54, 3\}$ (1.54 is the value in Kurlat's Exercise 7.5) and read it.

Solution: